📓 Day 11 Notes ✏️

Moving the Try-On into the Real Project

★ Today = my try-on left the practice folder and became a real React component in our project, on my own branch. The shirt follows my body inside the real app now.

1 My Own Branch

```
cd Desktop\virtual_tryon
git checkout main
git pull                        → get my friend's merged cart
cd client
npm install
cd ..
git checkout -b feature/tryon
```

Start from the latest main, then make a new branch. My work stays separate from my friend's pages until it is tested and merged.

2 The Reference Folder Trick

Antigravity can only read files INSIDE the project. So I made a temporary folder and copied my working Day 10 files into it (with File Explorer: Ctrl + C, then Ctrl + V):

```
mkdir reference
reference/index.html   ← my Day 10 practice code
reference/shirt.png
```

Then I told the agent to READ that file and port it. That is more reliable than describing the feature in words. The folder is only temporary, and I delete it at the end.

3 What I Asked Antigravity to Build

- TryOn.jsx in client/src/tryon, with a prop clothingImage (a URL) and an optional prop showControls for the sliders
- MediaPipe installed with npm (instead of the CDN link)
- The shirt copied to client/public/shirts/shirt.png
- A test page at /tryon-test, with a tiny change to App.jsx
- A "Loading model..." message, and a friendly error if the camera is blocked

- Same behaviour as Day 10: mirror view, smoothing 0.35, hiding the shirt, torso fallback. Slider defaults: width 2, height 1.4, offset 0.12
- Do NOT touch my friend's pages, the cart or the server

4 New Ideas

- Props: the component takes the clothing image as a prop. Each product page can send in its own shirt later.
- useRef: holds things that are not shown on screen, like the video, the canvas and the camera stream.
- Effect cleanup: a function that runs when someone LEAVES the page.
- StrictMode: in development, React runs effects twice to catch missing cleanup.

```
// simplified idea
useEffect(() => {
  // start the camera + the drawing loop

  return () => {
    // stop the camera tracks
    // cancelAnimationFrame
    // close the pose landmarker
  }
}, [])
```

Without cleanup, the camera light would stay on after leaving the page, and the loop would keep running in the background.

5 Bug 1: Code Looked Like One Line

In the editor, the whole file looked like one giant line, and the line-number column showed only a "2".

- Danger: on a single line, everything after a // counts as a COMMENT, so the rest of the code would silently stop working.
- How to check: look at the line numbers. A real file has a number on every line.
- Fix: ask the agent to rewrite the file with normal formatting, and change nothing else.

6 Bug 2: The Black Box

The page loaded and the camera light was on, but the video area was black.

```
// the problem: the same video in different branches
{status === "loading" && <video />}
{status === "running" && <video />}
```

When the status changed, React threw away the old video and made a new one. But my camera stream and my drawing were still attached to the OLD one, which was no longer on the page.

Fix: render the video and canvas ONCE, in the same place, for every status. Show "Loading..." and the error messages as overlays on top. After this, the shirt followed my body.

7 Checks and Saving

- Go to /tryon-test, allow the camera, stand back so both shoulders show
- Click Home. The camera light and the browser's camera icon should turn off
- Delete the temporary folder: rmdir /s /q reference
- git status should show only my files: TryOn.jsx, the small App.jsx change, the shirt image and the package files. Nothing from my friend's pages or the server.

```
git add .
git commit -m "Add try-on component and test page"
git push -u origin feature/tryon
```

Then make a pull request into main, like we did for my friend's work.

💡 Big takeaway:

A feature built in a practice file can move into the real app, if I give the agent the working code to read. Props make it reusable. Cleanup stops the camera when I leave the page. Never render the same video in different if-branches, because React will replace it. Next (Day 12): connect it to my friend's product page with a Try it on button. Products will need clothing pictures with transparent backgrounds.